

Digital Communications

Pulse Modulation



Introduction

- Topics in this class represent the transition between analog communication and digital communication
- In analog modulations, a parameter of the sine wave (carrier) was varied as a function of the modulating signal
- In pulse modulations, one of the parameters of the transmitted pulse train varies as a function of the modulating signal
- There are 2 types of pulse modulations:
 - Analog Pulse Modulation
 - Digital Pulse Modulation



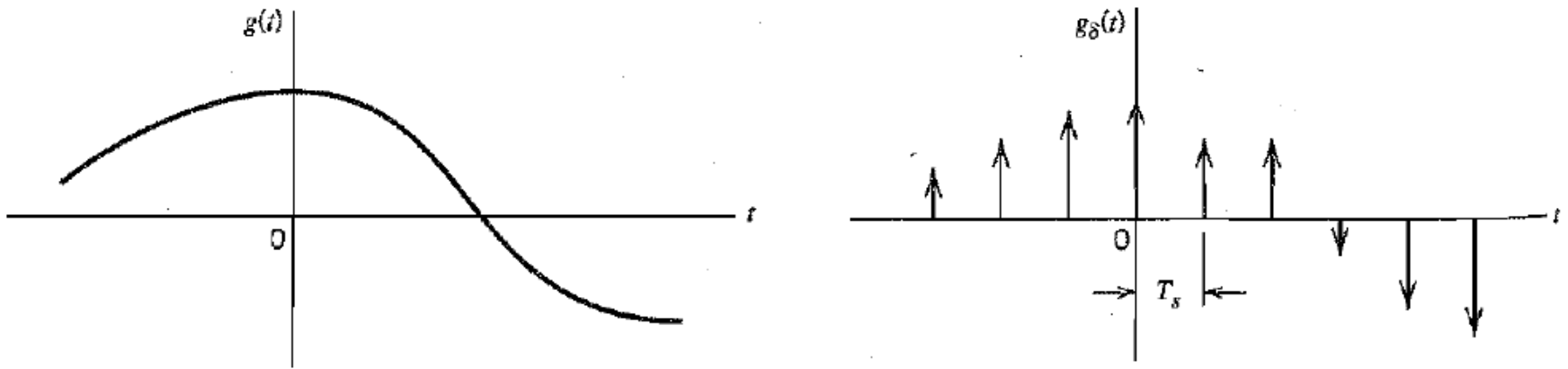
Introduction

- In the analog variant, a periodic pulse train is used as a carrier and one of the magnitudes (e.g. amplitude, duration, position) of each pulse varies continuously with the sampled value of the modulating signal
 - Information is transmitted in analog form but at discrete instants
- In the digital variant, the modulating signal is represented, in a discrete way, either in time or in amplitude
 - Allows the transmission in digital form of a sequence of coded pulses



Sampling

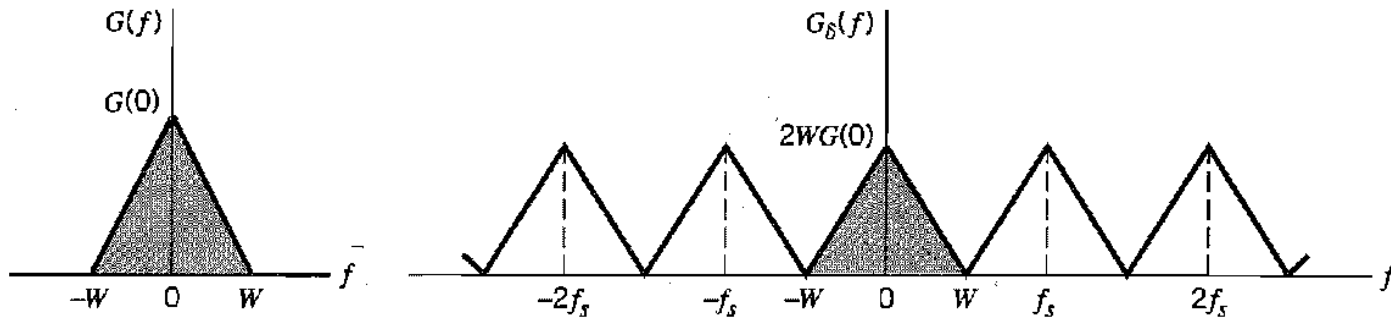
- Through a sampling process, an analog signal is converted into a sequence of samples, normally spaced evenly in time



- If the modulating signal band is limited [$G(f)=0, |f|>W$] and if we choose the sampling period $T_s = 1 / 2W$

Sampling

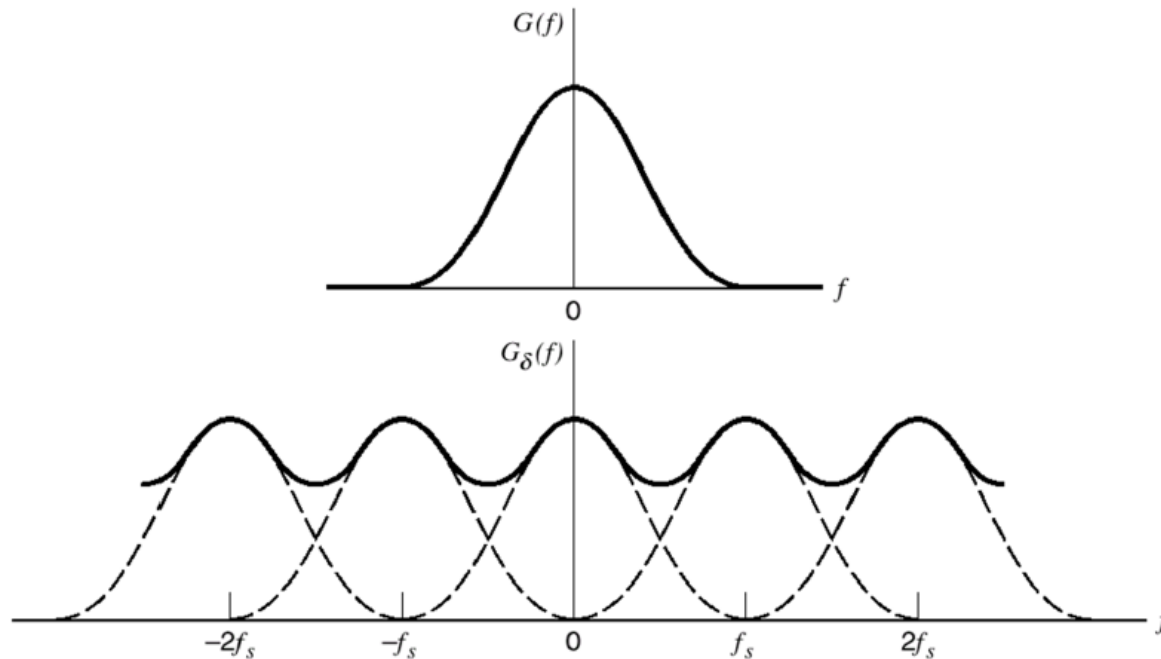
- The spectrum of the sampled signal is:



- Sampling Theorem (*Nyquist*):
 - A limited-band, finite-energy signal that does not have frequency components above W is completely described by specifying its values at time instants separated by $1/2W$
 - A limited-band, finite-energy signal that has no frequency components above W can be completely recovered by knowing its samples at a rate of $2W$ per second

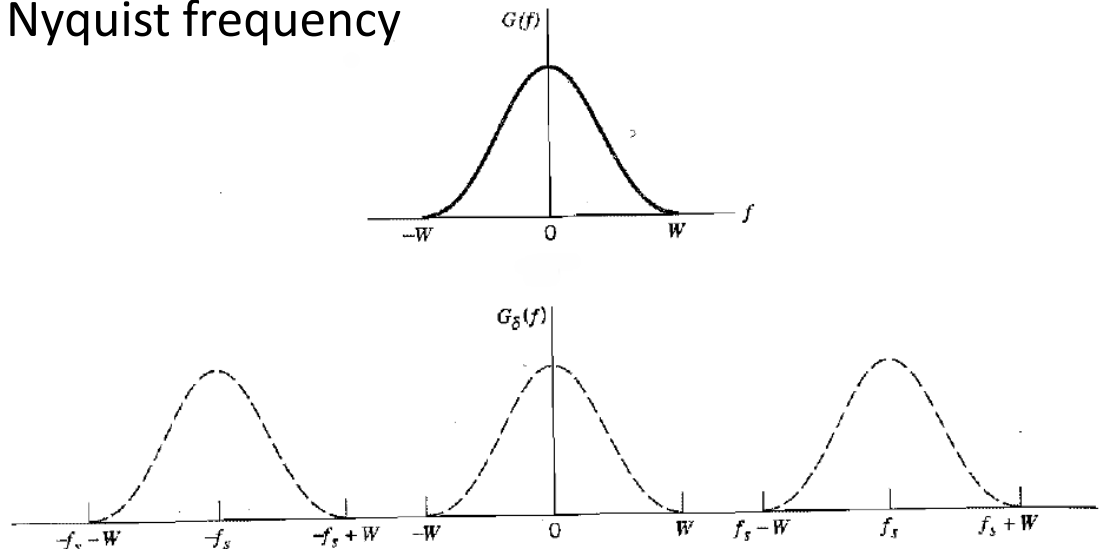
Sampling

- If the modulating signal is not band-limited or Nyquist's theorem is not respected, the phenomenon of under-sampling will occur and the sampling process will produce the aliasing effect



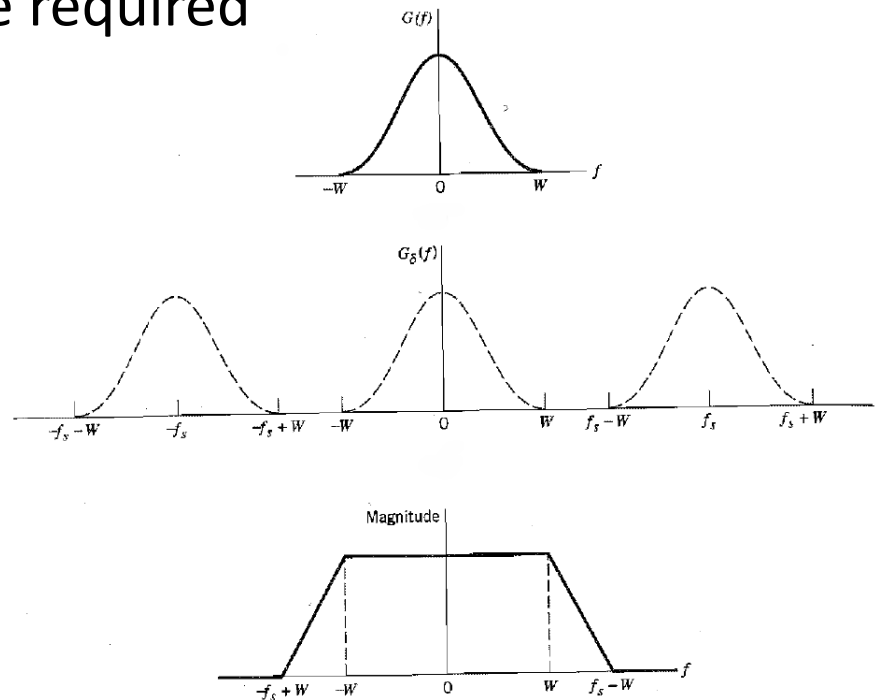
Sampling

- In order to combat the aliasing phenomenon, the following method is used:
 - Before sampling the signal, a low-pass filter is used in order to eliminate frequency components above twice the sampling frequency, as they are not essential in signal transmission
 - At the output of the filter, the signal is sampled at a frequency slightly higher than the Nyquist frequency



Sampling

- Thus, a low-pass filter with an extended bandwidth from W to $f_s - W$ will be required



- Thus, the filter has a well-defined transition band and is physically realizable

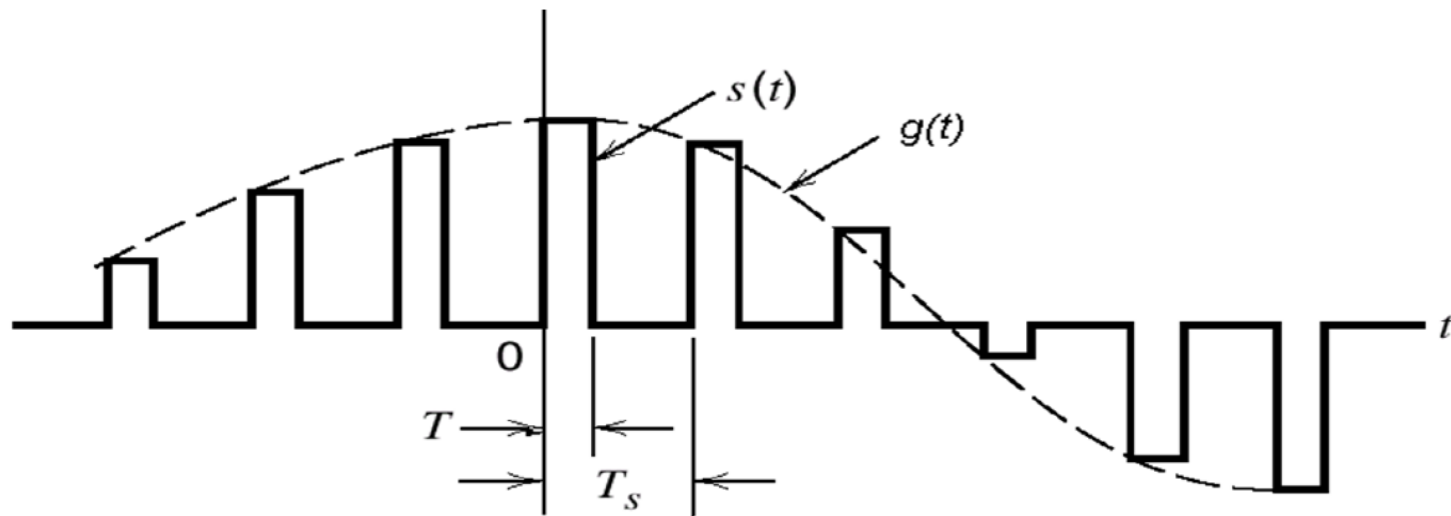
Pulse Amplitude Modulation

- The amplitudes of regularly spaced pulses vary as a function of the sampled values of the modulating signal
- PAM is very similar to natural sampling, where a continuous signal is multiplied by a train of periodic rectangular pulses
 - The difference is that, in the case of PAM, the impulse threshold is kept flat, while in natural sampling it varies with the modulating signal
- Two operations are required to produce a PAM signal:
 - Instantaneous sampling of the modulating signal $m(t)$ every T_s seconds, where the sampling rate $f_s = 1/T_s$ is chosen according to Nyquist's theorem
 - The duration of each sample is extended to a constant value T



Pulse Amplitude Modulation

- The electronic circuit that performs these two operations is called ***sample-and-hold***
- The idea of widening the pulses is in the sense of “saving” bandwidth, since the pulse width is inversely proportional to the LB



Pulse Amplitude Modulation

- The use of flat pulses resulting from signal sampling introduces amplitude distortion
- Although this distortion can be corrected, the transmission of the PAM signal imposes quite demanding requirements on the transmission channel in terms of phase and amplitude response, due to the reduced size of the transmitted pulses
- The performance of a PAM signal in terms of noise is always lower than that of a signal transmitted in baseband
- For transmissions over long distances, PAM modulation can be used only as a means of processing the message in order to perform TDM multiplexing, after which another modulation form will be used

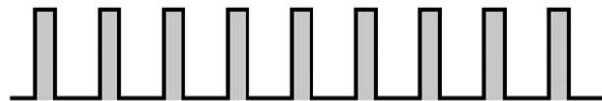
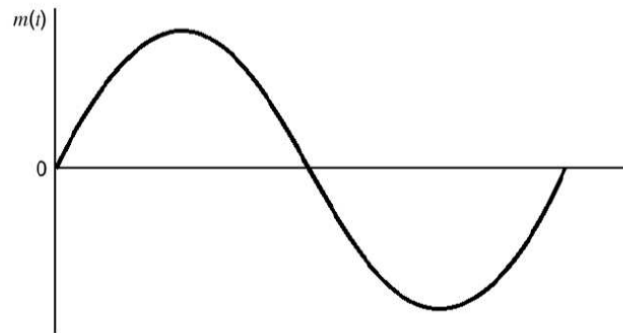


Other Forms of Pulse Modulation

- It is possible to take advantage of the increased transmission bandwidth consumed by pulses to increase system performance in terms of noise
- This task is achieved by representing the modulating signal in a property of the pulses other than amplitude
- There are 2 other forms of impulse modulation:
 - ***Pulse-Duration Modulation*** (PDM), also referred to as PWM (*Pulse Width Modulation*) where samples of the modulating signal are used to vary the duration of the carrier pulses
 - ***Pulse-Position Modulation*** (PPM), where the position of a pulse relative to its unmodulated position varies with the modulating signal



PDM / PPM



Impulse train



PDM Modulation



PPM Modulation

PDM / PPM

- In the case of PDM modulation, large pulses imply a high energy consumption
- Subtracting the part of power not used to transmit the information to the PDM signal, leaving only the transitions, the PPM signal is obtained
- Therefore, the PPM signal is more efficient than the PDM signal
- Additive noise affects the PPM transmission system performance, since, although the noise does not affect the signal, if the pulses are rectangular (since the information is in the position and not in the amplitude) the received signal, because the channel bandwidth is not infinite, causes received pulses to have a finite rise time



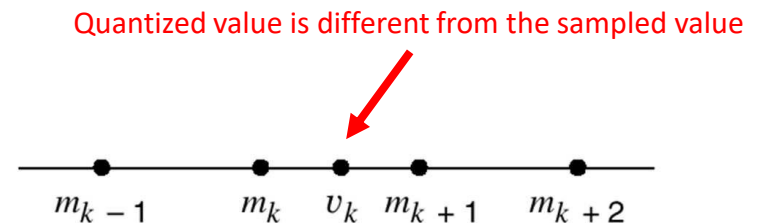
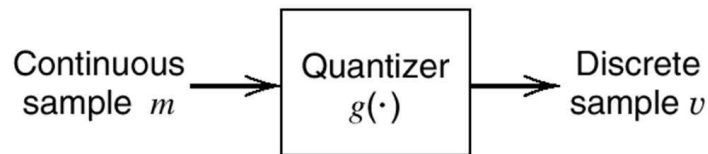
Quantization Process

- The quantization operation consists in the quantification of the PAM signal in q levels
- Considering the analog signal a normalized voltage signal such that: $|x(t)| \leq 1V$
- If the quantization is uniform, the voltage range of $2V$ (the maximum value of the peak-to-peak signal amplitude) is divided into q equally spaced levels. The separation between levels is:
 - $\Delta = 2 / q$
- The discrete levels are :
 - $\pm 1/q, \pm 3/q, \pm \dots, \pm (q-1)/q$
- The encoder then encodes each discrete level into a digital code word



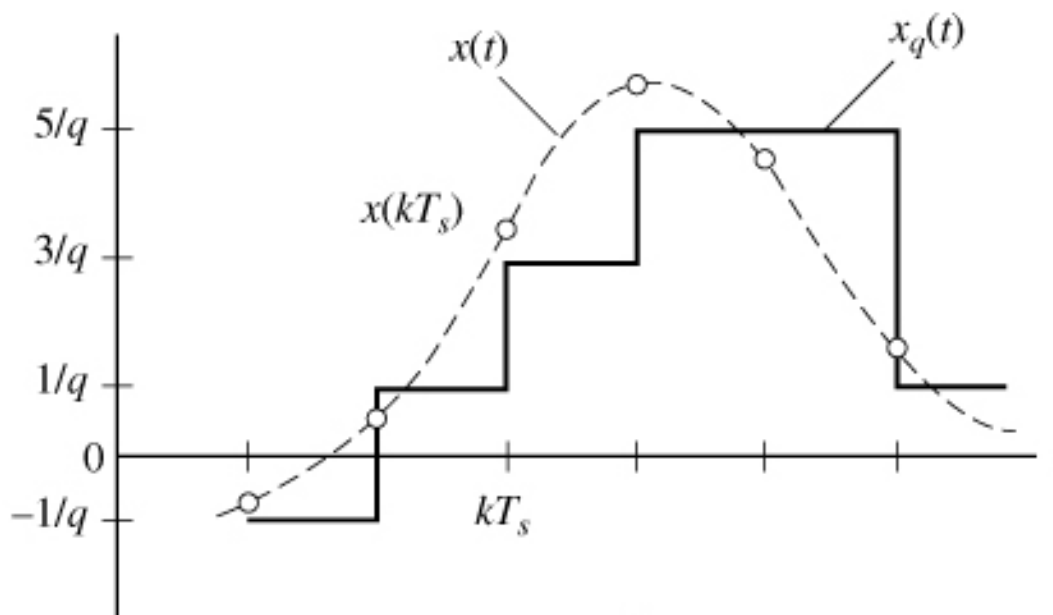
Quantization Process

- Operations involved:
 - Sampling according to Nyquist's theorem
 - Quantization: The signal is quantized to the nearest level
 - In this operation, the quantization error is made
 - Each quantization level is encoded using v bits, $q = 2^v$
 - Each quantized sample corresponds to an equivalent set of bits



Quantization Process

- Quantization error:
 - The quantization error of the k symbol, defined by the random variable ε_k , is the difference between the sampled signal $x(kT_s)$ and the quantized signal $x_q(kT_s)$
 - $\varepsilon_k = x_q(kT_s) - x(kT_s)$



Quantization Process

- Quantization is uniform or linear when the amplitude difference between two consecutive levels is constant
 - $\Delta = \text{const}$
- If the amplitudes to be quantized are evenly distributed between $[-1V, +1V]$, the distribution of ε_k is also uniform between $[-\Delta/2, +\Delta/2]$, that is, its *pdf* is given by :

$$f(\varepsilon_K) = \begin{cases} \frac{1}{\Delta} & \dots\dots -\frac{\Delta}{2} \leq \varepsilon_K \leq \frac{\Delta}{2} \\ 0 & \dots\dots \text{Other values} \end{cases}$$

- The power of the quantization noise is given by :

$$\sigma_q^2 = \overline{\varepsilon_K^2} = \int_{-\infty}^{\infty} f(\varepsilon_k) \varepsilon_k^2 d\varepsilon_k = \int_{-\Delta/2}^{\Delta/2} \frac{\varepsilon_k^2}{\Delta} d\varepsilon_k = \frac{\Delta^2}{12}$$

Quantization Process

- The previous result indicates that the noise power increases when the spacing between levels increases
 - what was intuitively expected
- One measure of the influence of quantization error on the performance of a PCM system is the relationship between signal strength and quantization noise power:
 - SNR – *Signal-to-Noise Ratio*)
 - $\text{SNR} = S_x / \sigma_q^2 \leq 1.8 + 6v$ --- (Simon Haykin, pag. 197)
- The expression indicates that the signal-to-noise ratio is proportional to the number of bits used to encode each sample and increases by 6 dB for each bit



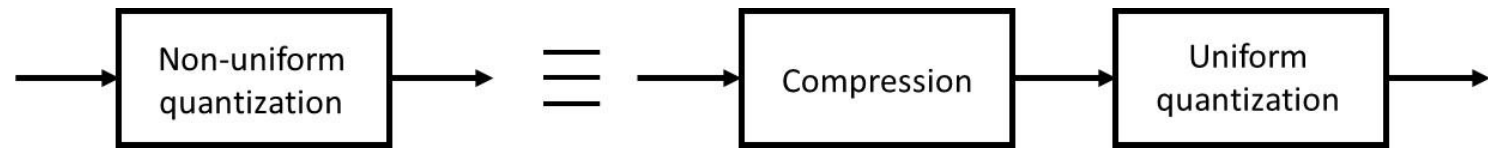
Non-uniform quantization

- Consider a signal that has low amplitudes most of the time
- With uniform quantization the power of the quantization noise is always constant (it just depends on the step!)
 - ⇒ as the signal strength is small the SNR ratio will be low most of the time
- Solution: for low signal amplitudes we should use steps closer to each other
- The step value is no longer constant: we have non-uniform quantization
- With non-uniform quantization the quantization signal-to-noise ratio becomes more independent of the signal level



Non-uniform quantization

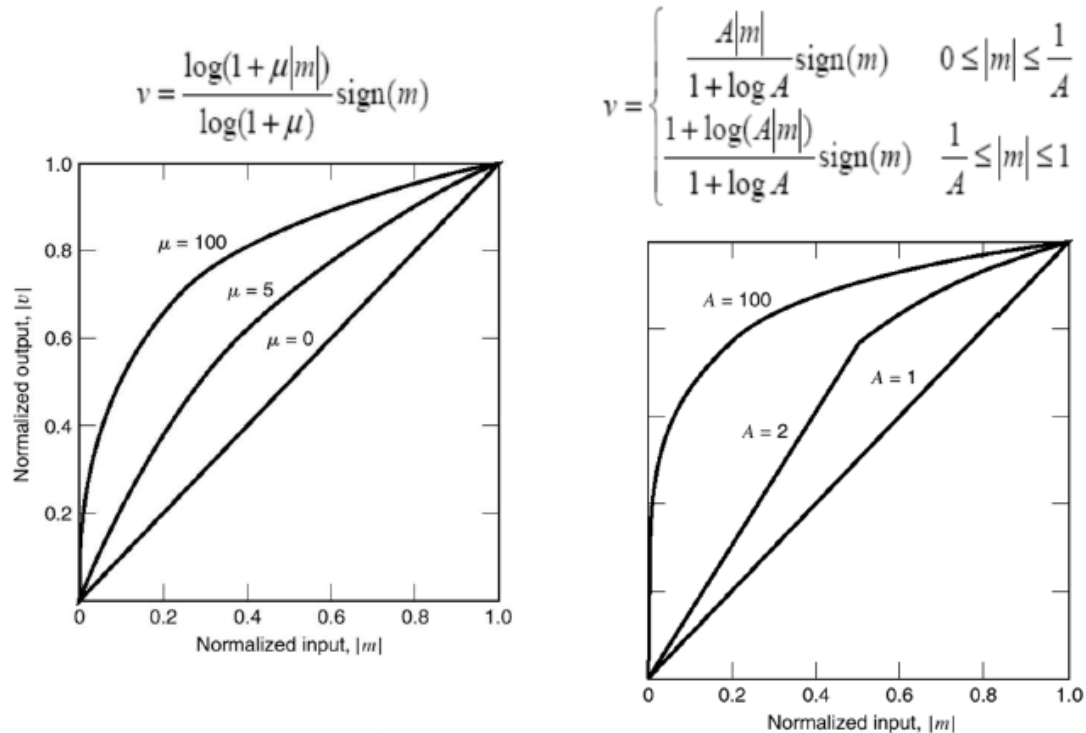
- Non-uniform quantization corresponds to two consecutive operations:
 - nonlinear compression and uniform quantization



- An “expander” must be used on the receiver
- Two “laws” of compression are used:
 - μ Law in North America and Japan
 - A Law in Europe
- The compression characteristics were determined from experimental studies performed with representative signals

Non-uniform quantization

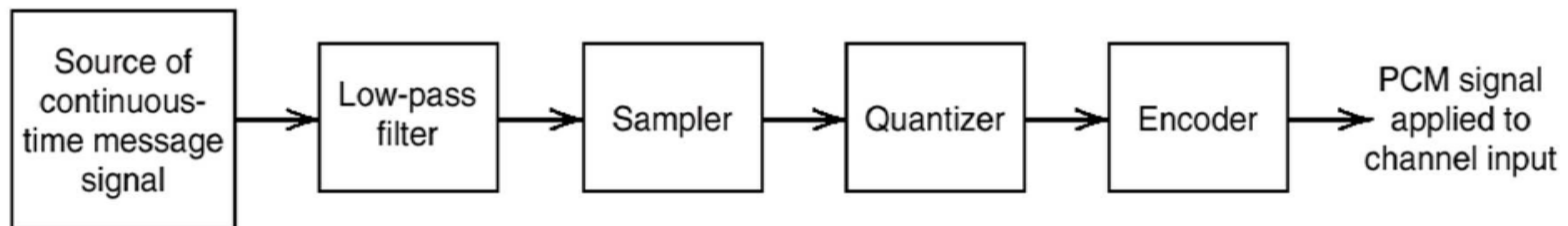
- Compression laws μ and A for voice signals:



- It is verified that, for low values of the signal strength, the use of non-uniform quantization allows saving four bits to obtain the same signal-to-noise ratio

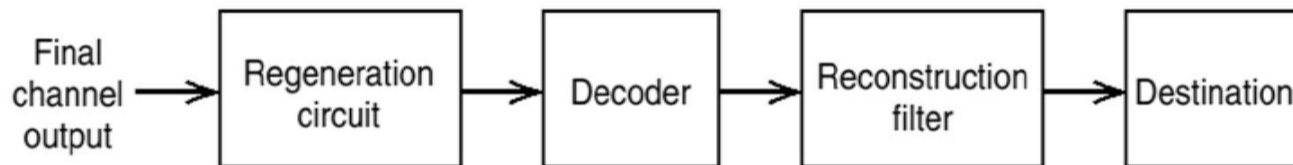
Pulse Code Modulation

- PCM modulation is the most basic form of digital impulse modulation
- A signal to be transmitted is represented by a sequence of coded pulses
- The signal to be transmitted is discretely represented either in amplitude or in time
- The basic operations carried out at the emitter are:
 - Sampling, Quantization and Coding



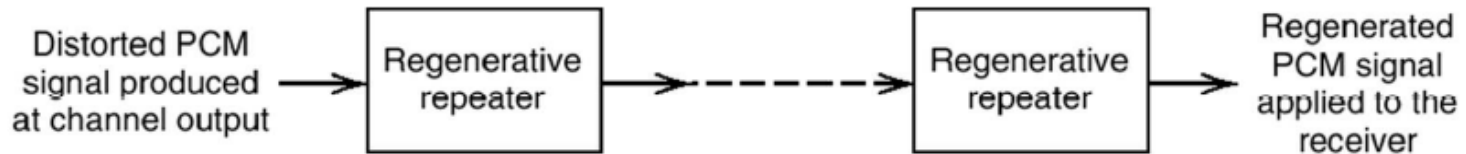
Pulse Code Modulation

- The low-pass filter is used to prevent aliasing
- The quantization and encoding operations are normally carried out in the same circuit (ADC – Analog to Digital Converter)
- The basic operations on the receiver are:
 - Signal Regeneration, Decoding and Sample Sequence Reconstruction



Pulse Code Modulation

- Along the way, the signal is regenerated in order to keep it within the desired levels in order to be correctly interpreted



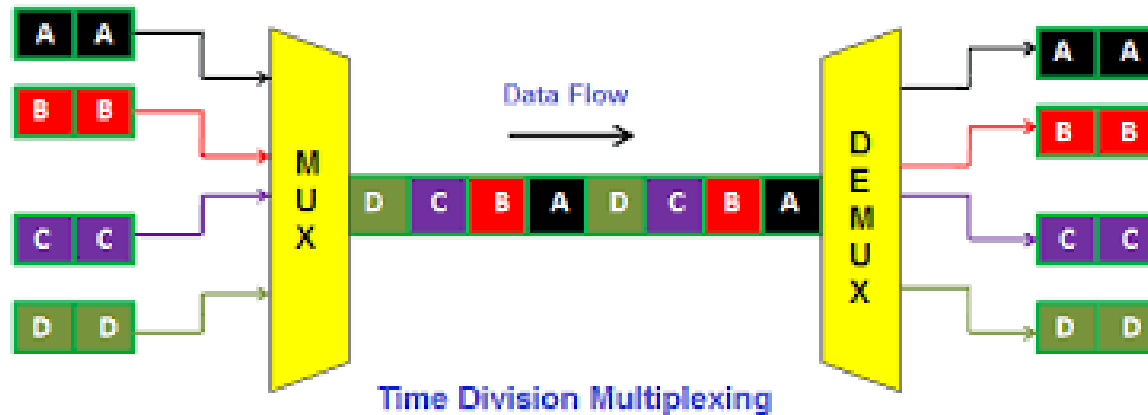
- The Coding process consists of converting the discrete values, resulting from the sampling and quantization processes, into a binary code of v bits
 - Thus, the value v corresponds to the number of bits per sample
 - It is possible to represent 2^v different values
 - For example, a sample quantized in one of 256 levels could be represented by an 8-bit codeword

Time Division Multiplexing

- Time division multiplexing is primarily used for digital signals, but can be applied to analog multiplexing in which two or more signals or streams of bits are transferred simultaneously on a single transmission medium
- The time domain is divided into several fixed-length time slots (time-slots), one for each channel
- A sample byte or data block from channel 1 is transmitted during time slot 1, channel 2 during time slot 2, etc
- A TDM frame consists of a time slot per channel plus a sync channel and sometimes an error correction channel
- After the last channel, the cycle starts again with a new frame, starting with the second sample, byte or data block from channel 1, etc

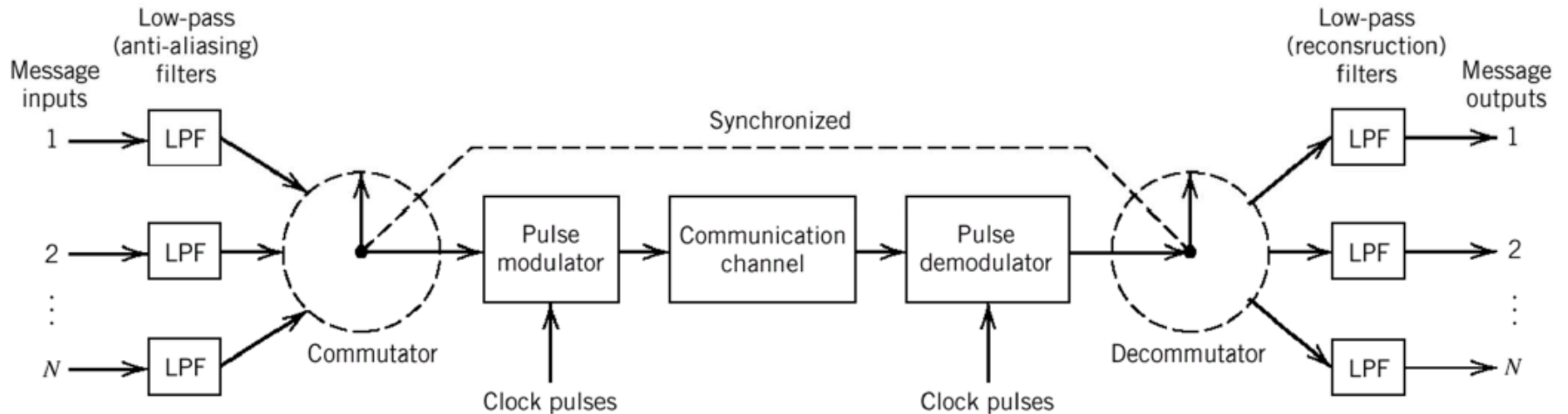


Time Division Multiplexing



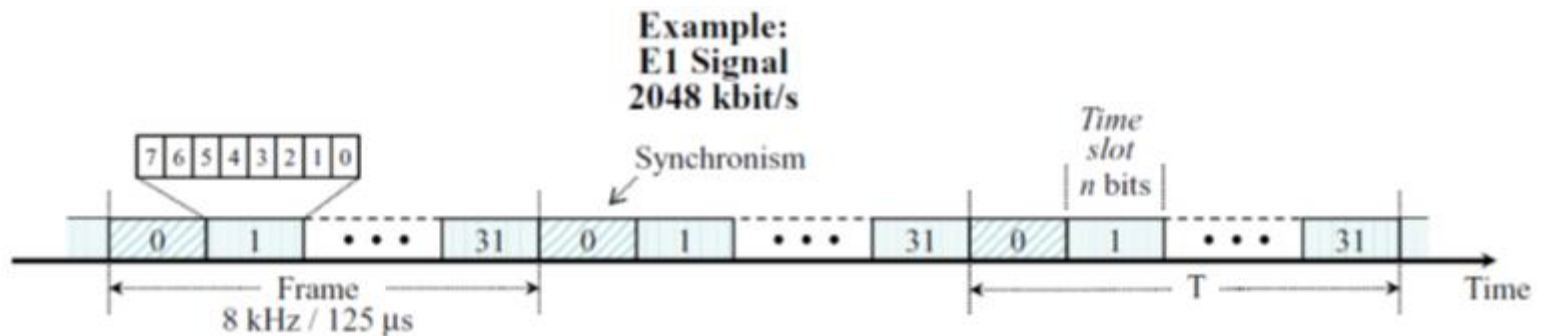
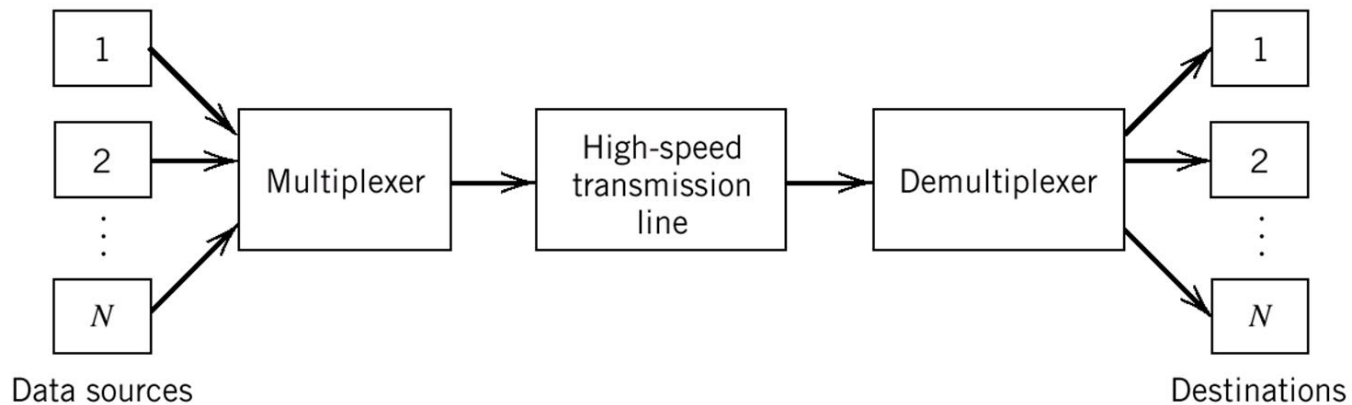
Time Division Multiplexing

- Block diagram of a TDM multiplexing system



Time Division Multiplexing

- TDM Multiplexing / Demultiplexing



Pulse Code Modulation

- Advantages of PCM Modulation:
 - Robustness to Interference and noise in the transmission channel
 - Efficient regeneration of the encoded signal along the entire transmission path
 - Through a uniform information format, it allows the integration of several digitally encoded baseband signals in a single transmission medium
 - Each multiplexed signal can be removed or inserted, along the way, in a TDM multiplexing system
 - Although the use of PCM coding involves some relatively complex operations, its implementation is now done in a massive way through commercially available components using VLSI (*Very Large Scale Integration*) technology
 - PCM technology is the most used in the digital transmission of analog signals



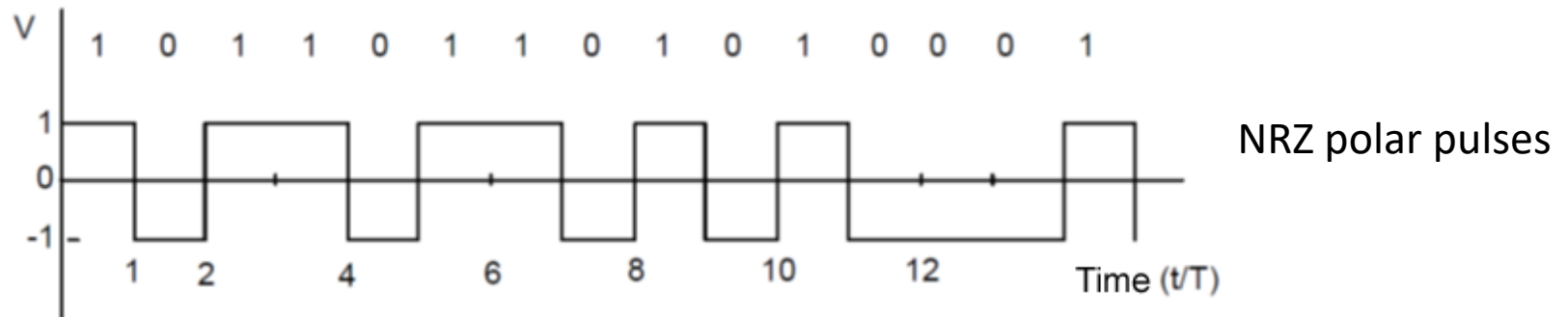
Line Codes

- Line codes are used to adapt the characteristics of information symbols to the characteristics of the transmission medium
- How to represent the sequence 1 0 1 1 0 1 1 0 1 0 1 0 0 0 1, where each bit lasts for T seconds?
- One possibility is to assign bits “1” to a rectangular pulse of positive polarity and duration T , and to assign bits “0” to a null pulse (i.e. no pulse):



Line Codes

- It is also possible to represent the sequence of bits as follows, using **NRZ polar pulses**:



- There are many other line codes. The choice depends on the features we want

Line Codes - Requirements

- Depending on the application, some of the following characteristics are desirable in line encoding:
 - Null DC component:
 - This feature is convenient if there are transformers in the system, as they block the DC component of the input signals
 - Sufficient timing information for clock recovery at the receiver
 - Low-bandwidth frequency spectrum situated at low frequencies
 - To minimize the influence of crosstalk, thermal noise, impulse noise, radio interference and cable attenuation
 - “Transparency” (i.e. suitability) for all types of messages
 - Unique decoding (unambiguous)

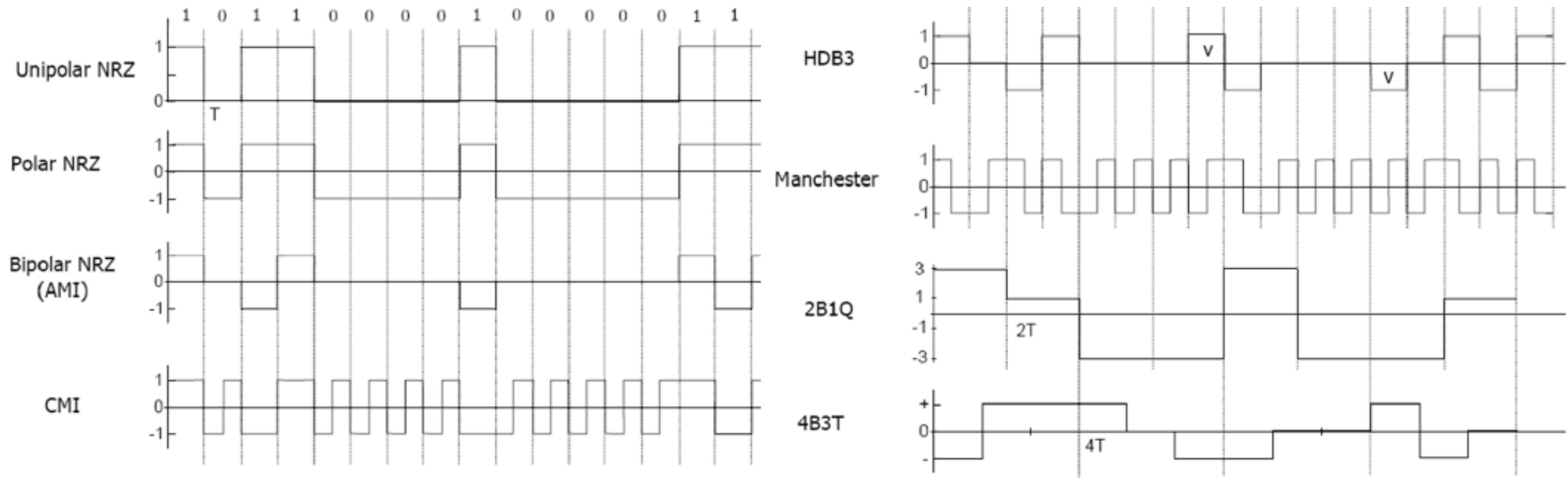
Line Codes - Requirements

- Line Code Characteristics (continued)
 - High immunity to additive disorders :
 - This feature favors a binary line signal because it combines good quality detection at the receiver with ease of generation at the emitter
 - Error correction capability
 - Ease of equalization of channel characteristics, if necessary
 - Linear relationship between encoded and decoded signals



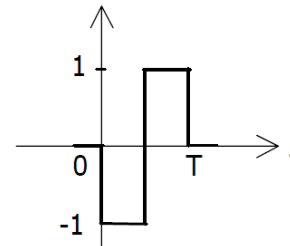
Line Codes

- Examples:



Line Codes

- In AMI encoding, an input sequence with many consecutive “0” bits gives a long pulseless output
- If there are no impulses, there are no transitions, or zero crossings, which is not desirable from the point of view of clock recovery
- Two variants that ensure that there are no long gaps without transitions are the HDB3 and CMI codes:
 - CMI (*Coded Mark Inversion*)
 - Bit “1” is represented exactly as in AMI: a pulse of duration T seconds of alternating polarity
 - The “0” bit is represented by the pulse



Line Codes

- HDB3 (*High Density Bipolar with 3 “0” maximum*)
 - In the absence of sequences of more than three consecutive zeros, this code is identical to the AMI code
 - If four or more consecutive zeros appear, extra pulses are introduced – the *V* pulses and the *B* pulses
 - In a sequence of four or more zeros, the fourth zero is represented by a pulse with a polarity such that it violates the AMI coding rule, i.e., it has the same polarity as the previous “1”. This impulse is called impulse “*V*” (Violation)
 - These extra impulses do not guarantee a similar number of positive and negative impulses
 - We have to introduce so-called “*B*” (Balance) impulses in accordance with AMI rules:
 - If a *V* pulse has the same polarity as the previous *V* pulse, a *B* pulse is inserted into the first of the four “0” bits
 - If the *V* pulses have opposite polarities, it is not necessary to precede the second of a *B* pulse
 - The encoder has to store each input bit until three subsequent bits have arrived



Line Codes

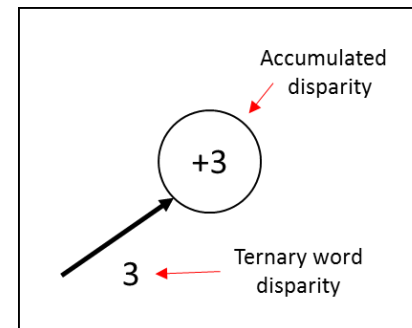
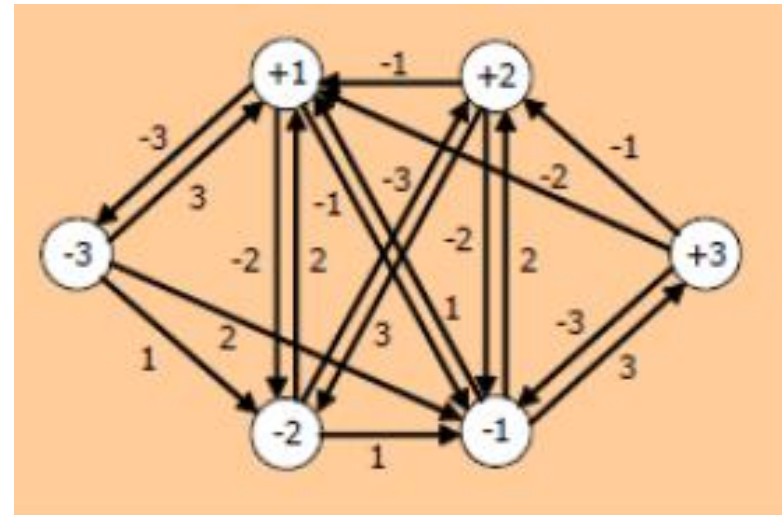
- 4B3T
 - In this code, a block of 4 binary digits is converted into a block of 3 ternary digits
 - Of the $3^3 = 27$ possible ternary output blocks, only 16 are used, corresponding to the $2^4 = 16$ possible binary input blocks
 - The difference between the number of positive (“+”) and negative (“-”) levels is stored as encoding takes place. This difference is called cumulative disparity
 - The choice of ternary blocks is made according to the signal's past history and follows a state transition diagram or a table
 - The baud rate (number of symbols/sec) is 75% of the original bit rate



Line Codes

- 4B3T (Continued)

Binary Word	Ternary Word		Disparity
	Positive Mode	Negative Mode	
0000	0-+	0-+	0
0001	-+0	-+0	0
0010	-0+	-0+	0
1000	0+-	0+-	0
1001	+ -0	+ -0	0
1010	+0-	+0-	0
0011	+++	+-	1
1011	+00	-00	1
0101	0+0	0-0	1
0110	00+	00-	1
0111	-++	+--	1
1110	++-	--+	1
1100	+0+	-0-	2
1101	++0	--0	2
0100	0++	0--	2
1111	+++	---	3



Line Codes

- Spectral Power Densities

Unipolar NRZ $S(f) = \frac{A^2 T}{4} \text{sinc}^2(fT) \left[1 + \frac{1}{T} \delta(f) \right]$

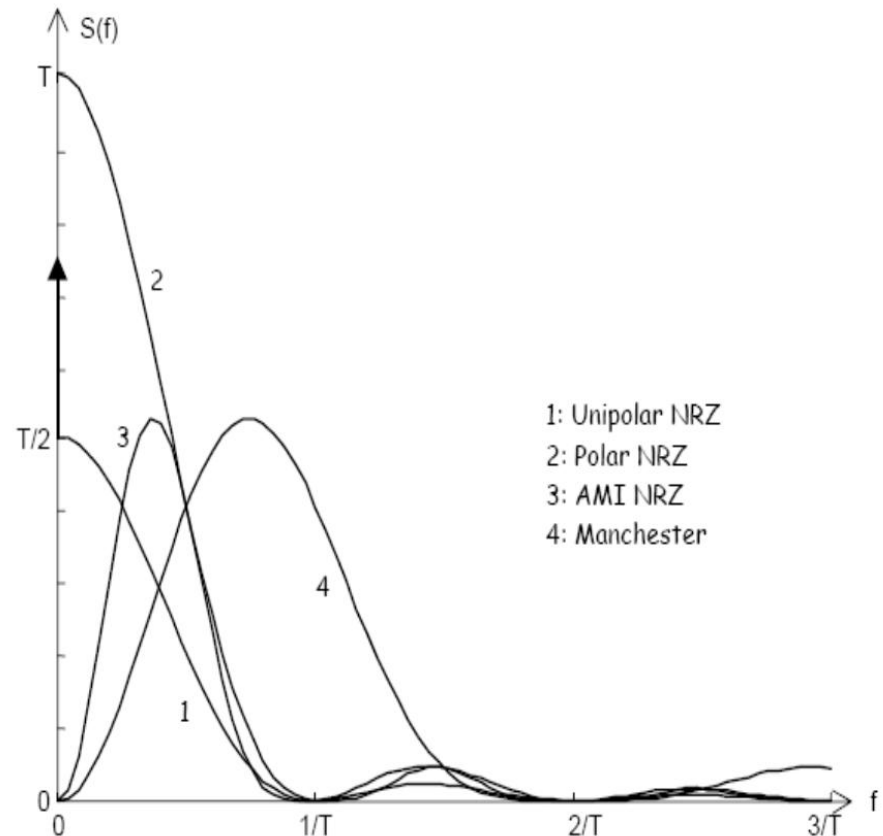
Polar NRZ $S(f) = A^2 T \text{sinc}^2(fT)$

Unipolar RZ $S(f) = \frac{A^2 T}{16} \text{sinc}^2\left(\frac{fT}{2}\right) \left[1 + \frac{1}{T} \sum_{n=-\infty}^{\infty} \delta\left(f - \frac{n}{T}\right) \right]$

AMI NRZ $S(f) = A^2 T \text{sinc}^2(fT) \text{sen}^2(\pi f T)$

AMI RZ $S(f) = \frac{A^2 T}{4} \text{sinc}^2\left(\frac{fT}{2}\right) \text{sen}^2(\pi f T)$

Manchester $S(f) = A^2 T \text{sinc}^2\left(\frac{fT}{2}\right) \text{sen}^2\left(\frac{\pi f T}{2}\right)$



Line Codes – where they are used?

- AMI
 - First to be used in copper symmetrical pair systems. Replaced with modified AMI codes
- HDB3
 - ITU-T G.703 standard for PCM systems multiplexed at 2, 8 and 34 Mbits/s
- CMI
 - ITU-T G.703 Standard for Multiplexed PCM at 140 Mbits/s
- Biphasic code (Manchester)
 - Distribution of clock signals in VLSI circuits
 - Magnetic recording
 - Ethernet Local Area Networks
 - RDS system (“Radio Data System”) in FM broadcasting



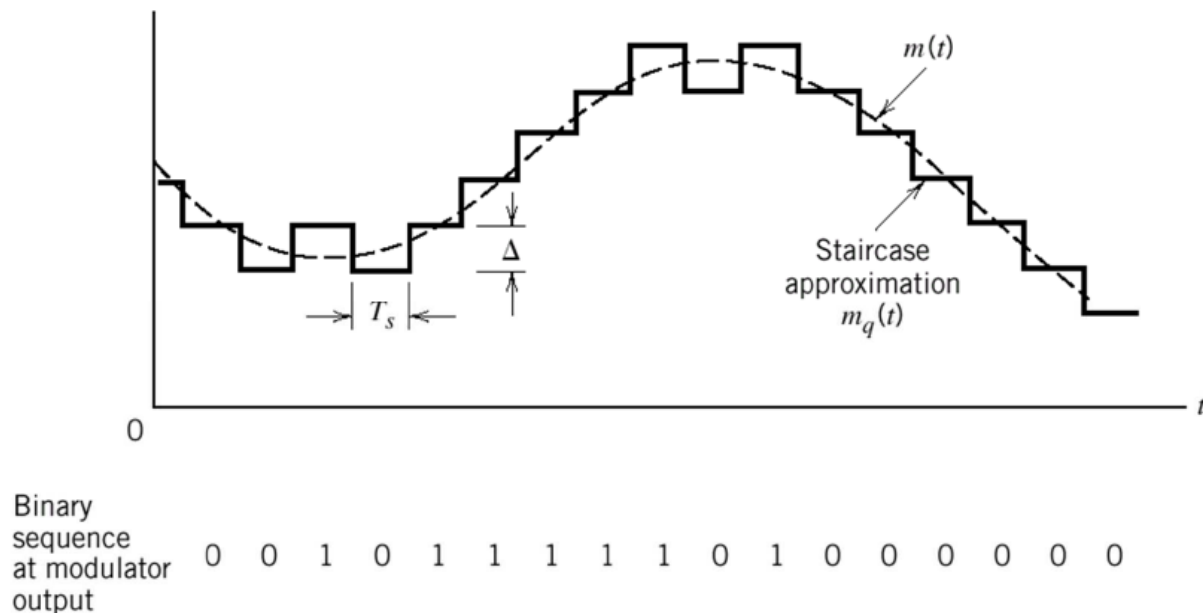
Line Codes – where they are used?

- 4B3T
 - High capacity systems (long distance systems on coaxial cable at 34 and 140 Mbits/s)
- 2B1Q
 - Basic ISDN Access (160 kbits/s)
- nBmB (5B6B, etc)
 - High-capacity, high-bandwidth systems (fiber optics)



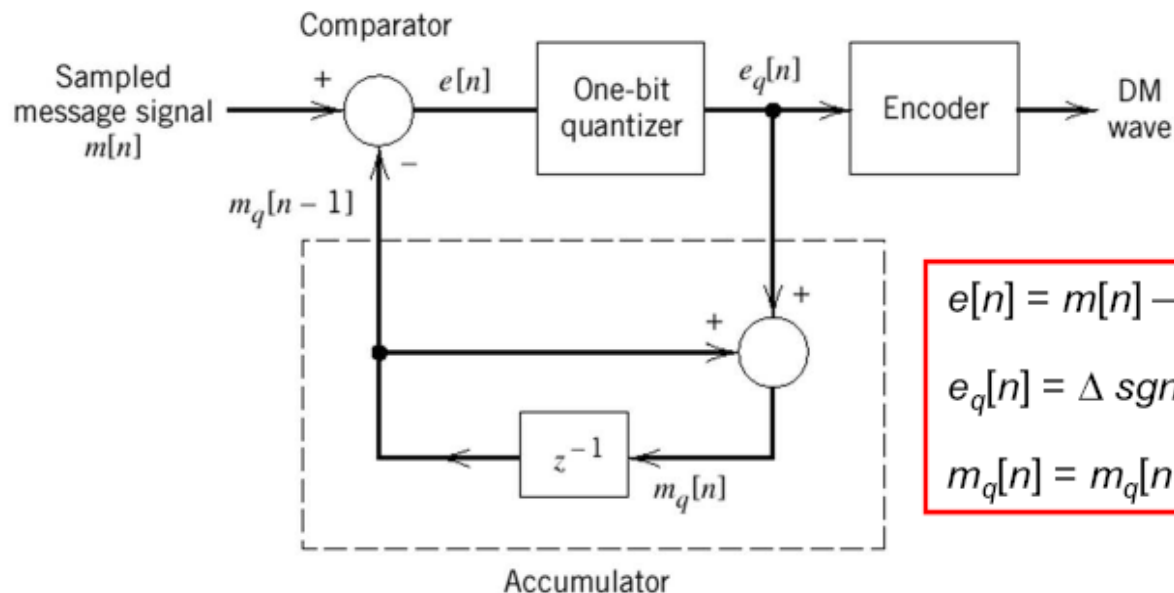
Delta Modulation

- In Delta modulation, the modulating signal is oversampled above the Nyquist frequency to increase the accuracy between quantizations of adjacent samples of the signal
- This strategy allows the use of 1-bit quantizers



Delta Modulation

- Delta modulation follows the input signal through successive approaches (in a ladder)
- The difference between the input signal and the signal quantized at the previous instant is quantized in the value $\pm\Delta$



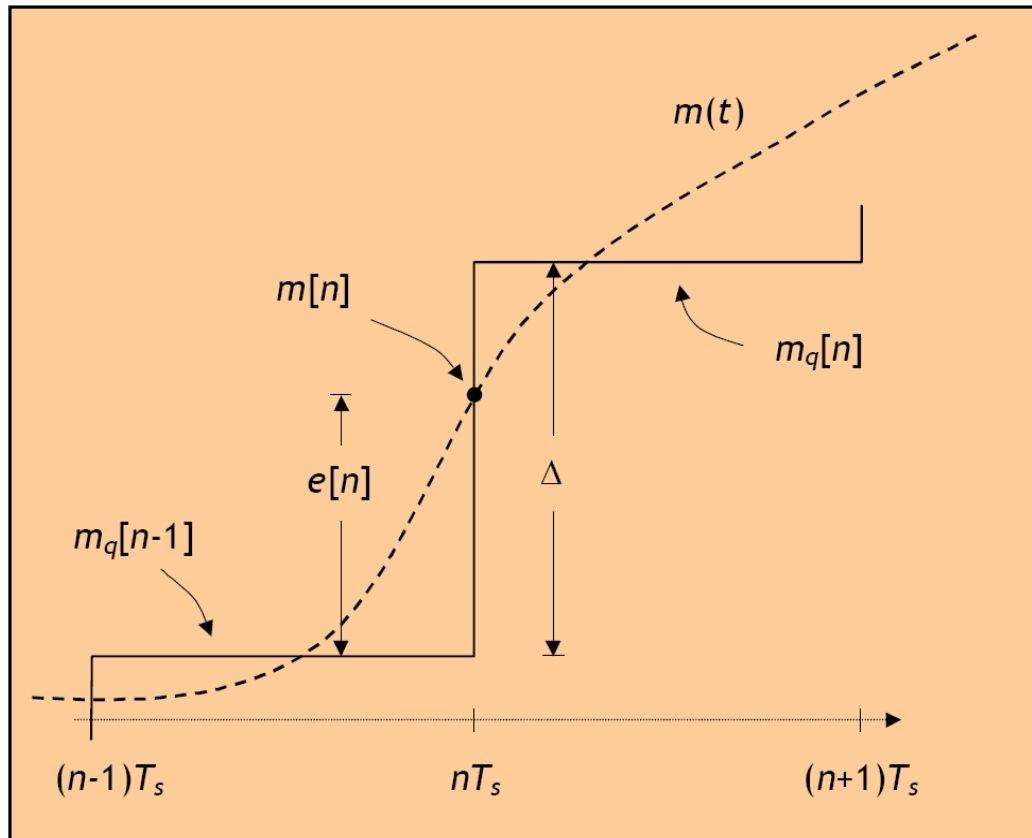
$$e[n] = m[n] - m_q[n - 1]$$

$$e_q[n] = \Delta \operatorname{sgn}(e[n])$$

$$m_q[n] = m_q[n - 1] + e_q[n]$$

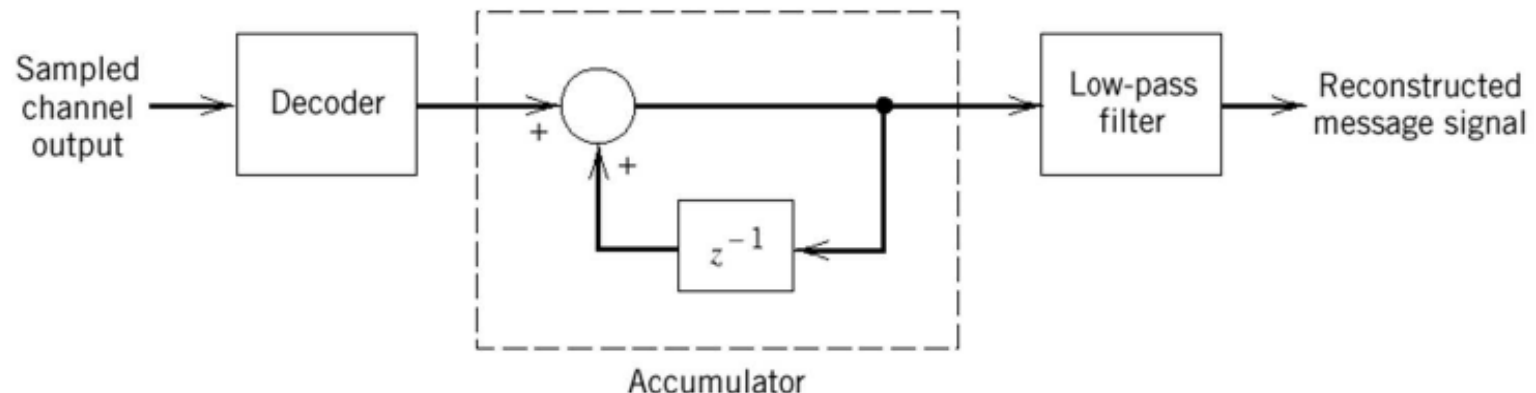
Delta Modulation

- Parameters involved in Delta Modulation



Delta Modulation

- At the receiver, the received signal is reconstructed at the output of the Decoder and Accumulator, in a similar way to the transmission
- The high frequency noise produced by the transitions (ladder) is filtered in the LPF

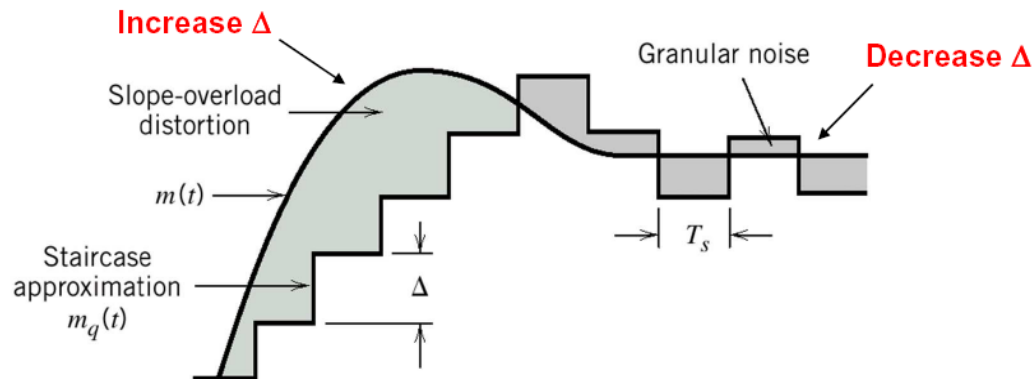


Delta Modulation

- Delta modulation is subject to 2 types of quantization noise:
 - *Slope overload*
 - May occur if the signal has very steep slopes: the stair approach cannot keep up with it
 - To cancel it, it is necessary that:

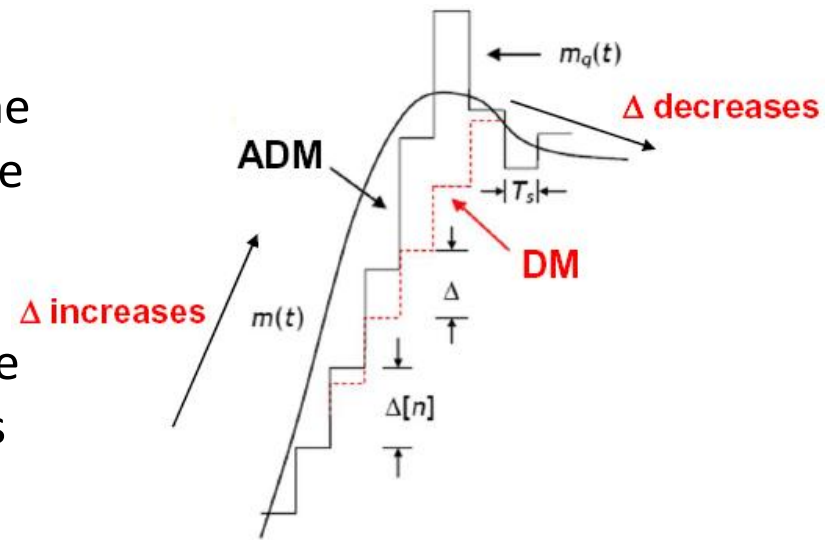
$$\frac{\Delta}{T_s} \geq \max \left| \frac{dm(t)}{dt} \right|$$

- Granular noise
 - Occurs in the “flat” zones of the signal



Adaptive Delta Modulation - ADM

- To avoid slope distortion it is necessary to increase Δ
- To reduce granular noise it is necessary to lower Δ
⇒ A compromise has to be reached!
- An alternative is to make the step variable through Adaptive Delta Modulation (ADM):
 - If successive errors $e_q[n]$ have the same polarity (“mode” of slope distortion) the step is increased
 - If successive errors $e_q[n]$ have opposite polarities (“granular mode”), the step is decreased



Other Forms of Pulse Modulation

- Differential Pulse-Code Modulation - DPCM
- Adaptative Differential Pulse-Code Modulation - ADPCM



Reading Material

- *Simon Haykin, Communication Systems, McMaster Univ, 5th Edition, 2009 (ISBN 978-047-16979-0-9)*
- *A. Bruce Carlson and Paul Crilly. Communication Systems – An Introduction to Signals and Noise, McGraw-Hill, 5th edition, 2009 (ISBN 978-007-338-040-7)*
- *Bernard Sklar and Fredric Harris, Digital Communications: Fundamentals and Applications, Pearson, 3rd Edition, 2020 (ISBN-13: 978-0134588568)*



Reading Material

